Let’s design the solution for this problem, referred to hereafter as DCC-Dapp (Data-intensive Computing Certificate Dapp).

11.3.2 Design choices

You have many choices of application to address this problem:

- A standalone mobile app independent of the central system
- A web or enterprise application integrated with the central student management system
- A blockchain-based decentralized application independent of the central system

The first two choices are traditional; the last one offers the independence of a mobile app and at the same time keeps track of the transactions on the immutable ledger of the blockchain. That’s the design you’ll explore in the following sections.

11.4 Analysis and design

Now let’s apply the design principles you’ve learned so far to design a solution for the problem. Using figure 11.2, DCC-concept, as a guideline, identify the roles, rules, assets, and functions. This step will enable you to think through the problem and design a contract diagram and its elements: